# Proofs and corrections of OEIS conjectures

29 papers in 7 families

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This collection settles conjectures recorded in 29 entries of the On-Line Encyclopedia of Integer Sequences (OEIS). Most of the sequences are coefficients of formal power series defined by functional equations, and most of the conjectures concern their residues modulo small integers. In 24 entries the conjectures treated are proved, and in A392210 an observation recorded in the entry is proved. In four entries (A380680, A396798, A396843 and A396844) a conjectured statement is false; the papers give counterexamples and prove the correct statements. Each paper is self-contained, and its main theorems are formalized in Lean 4 with Mathlib; the formal proofs use only Lean's standard axioms. All statements were also checked numerically against the published OEIS data.

The results were obtained in an AI research effort directed by Dev D. Goyal: ChatGPT (OpenAI, GPT-6 with Pro reasoning), working autonomously from Dev D. Goyal's prompts, found the proofs and wrote the Lean formalization, and Claude Opus 5.5, used with Claude Code, checked the work independently and wrote the papers in their present form. Dev D. Goyal has verified the results. The papers have not been peer reviewed.

## 1. The family $A_m(xA_m - mxA_m^2) = x^2$

Parity, residues modulo $m^2$ and binary-digit divisibility for the unique normalized solution of $A_m(xA_m - mxA_m^2) = x^2$, with corrected residue classifications for the two entries.

**A396843.** Odd coefficients, remainders modulo 9, and binary digits for the series $A(xA - 3xA^2) = x^2$

Every coefficient is odd; the exact residues modulo 9 (correcting the entry's second conjecture); $3^{s_2(n-1)}$ divides $a(n)$, where $s_2$ is the binary digit sum. *(proved; one statement corrected.)*

Paper · OEIS A396843 · Lean

**A396844.** Remainders modulo 16 for the series $A(xA - 4xA^2) = x^2$

The exact residues modulo 16; the conjecture fails first at $n = 17$. *(conjecture disproved; corrected statement proved.)*

Paper · OEIS A396844 · Lean

## 2. Consecutive iterates $F = x + F^{\circ r}\, F^{\circ(r+1)}$

For every integer $k$, the $k$-th iterate of $F_r$ modulo $2r$ is $x/(1-kx)$ when $r$ is odd, with an explicit correction term when $r$ is even.

**A396099.** The iterates of a self-referential power series modulo 4

All four conjectured congruences modulo 4 for the series and its iterates. *(proved.)*

Paper · OEIS A396099 · Lean

**A396798.** The iterates of a self-referential power series modulo 8

Seven of the eight conjectured congruences modulo 8; the remaining one holds from $n = 2$, not from $n = 1$ as stated. *(proved; one statement corrected.)*

Paper · OEIS A396798 · Lean

**A396807.** The iterates of a self-referential power series modulo 10

For every integer $k$, the coefficients of the $k$-th iterate satisfy $[x^n]\, A^{\circ k} \equiv k^{n-1} \pmod{10}$. *(proved.)*

Paper · OEIS A396807 · Lean

## 3. Scaled inverses $A(x - x\,A(qx)/q) = x$

Every coefficient is 1 modulo $q+1$; for even $q$, a coefficient is odd exactly when its index is a power of 2.

**A393856.** Congruences for the series $A$ with $A(x - x\,A(4x)/4) = x$
$a(n) \equiv 1 \pmod 5$, and $a(n)$ is odd exactly when $n$ is a power of 2. *(proved.)*
Paper · OEIS A393856 · Lean

**A393857.** Congruences for the series $A$ with $A(x - x\,A(5x)/5) = x$
$a(n) \equiv 1 \pmod 6$. *(proved.)*
Paper · OEIS A393857 · Lean

**A393858.** Congruences for the series $A$ with $A(x - x\,A(6x)/6) = x$
$a(n) \equiv 1 \pmod 7$, and $a(n)$ is odd exactly when $n$ is a power of 2. *(proved.)*
Paper · OEIS A393858 · Lean

**A393859.** Congruences for the series $A$ with $A(x - x\,A(7x)/7) = x$
$a(n) \equiv 1 \pmod 8$. *(proved.)*
Paper · OEIS A393859 · Lean

## 4. Theta series and theta-type equations

Series built from the Jacobi theta series $\theta_3$ or defined by theta-type sums, whose coefficients follow simple patterns modulo 4 or 16.

**A378580.** Remainders modulo 4 for the series with $A(x/A(x)) = \theta_3(x)$
$A(x/A(x)) = \theta_3$: for $n > 0$, $a(n) \equiv 2 \pmod 4$ if $n$ is a square, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A378580 · Lean

**A378581.** Remainders modulo 4 for the series with $A(x\,A(x)) = \theta_3(x)$
$A(x\,A(x)) = \theta_3$: the same square pattern. *(proved.)*
Paper · OEIS A378581 · Lean

**A380060.** Square indices and remainders modulo 4 in a theta-type series
For $n > 0$, $a(n) \equiv 2 \pmod 4$ if $n$ is a square, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A380060 · Lean

**A380680.** Twice the squares, not the squares: remainders modulo 4 in a theta-type series
The square conjecture is false ($a(1) = 4$, $a(2) = 6$); in fact $a(n) \equiv 2 \pmod 4$ exactly when $n = 2k^2$. *(conjecture disproved; corrected statement proved.)*
Paper · OEIS A380680 · Lean

**A355872.** Residues modulo 16 for a series defined by a bilateral theta-type equation
$a(n) \equiv 2(-1)^{n-1} \pmod{16}$ for all $n \geq 1$, which implies the three conjectured congruences modulo 4 and 8. *(proved.)*
Paper · OEIS A355872 · Lean

## 5. Catalan-type equations

Equations whose solutions reduce, modulo 2, 4 or 8, to expressions in the series $\sum_{n\geq 1} C_{n-1}x^n$ of Catalan numbers, whose $n$-th coefficient is odd exactly when $n$ is a power of 2.

**A393172.** The Catalan numbers inside the equation $A(x - x^2) = x + A(x)^2$
For $n > 1$, $a(n) \equiv 2 \pmod 4$ if $n = 2^k$, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A393172 · Lean

**A393171.** Remainders modulo 8 of a series built from the Catalan numbers
For $n > 2$, $a(n) \equiv 4 \pmod 8$ if $n = 2^k$ or $2^k + 1$, and $8 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A393171 · Lean

**A392210.** The odd-indexed terms of a nested square are even after the first term
$a(2n-1)$ is even for $n > 1$ (an elementary observation recorded in the entry). *(proved.)*
Paper · OEIS A392210 · Lean

**A389536.** Remainders modulo 4 for the equation $A(x^2 + 2x^3) = x\,A(x)$
For $n > 1$, $a(n) \equiv 2 \pmod 4$ if $n = 2^k + 1$, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A389536 · Lean

**A397588.** Odd coefficients at powers of two for $A = x + (xA^2)'$
$a(n)$ is odd exactly when $n$ is a power of 2, and $3 \mid a(n)$ for $n > 1$. *(proved.)*
Paper · OEIS A397588 · Lean

**A195194.** Parity of an infinitely nested series reversion
The infinitely nested reversion is well defined, and $a(n)$ is odd exactly when $n$ is a power of 2. *(proved.)*
Paper · OEIS A195194 · Lean

## 6. Reversions with an exact halving step

Equations whose coefficient recursion appears to divide by 2; the division is shown to be exact before reducing.

**A389537.** Congruences modulo 4 for the series with $A(A(x) - x) = A(x)^2$
For $n > 2$, $a(n) \equiv 2 \pmod 4$ if $n = 3 \cdot 2^k$, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A389537 · Lean

**A389542.** Congruences modulo 4 for the series with $A(A(x)^2 - x^2) = 4A(x)^3$
For $n > 1$, $a(n) \equiv 2 \pmod 4$ if $n = 2^k + 1$, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A389542 · Lean

**A389474.** A power series whose square is $x^2 + 2x^3$ modulo 9
$A^2 \equiv x^2 + 2x^3 \pmod 9$, and $A^{3k} \equiv A(x^3)^k \pmod 3$. *(proved.)*
Paper · OEIS A389474 · Lean

**A389475.** Two congruences modulo 3 for the series with $A(3x^2) = 2x^2 + A^2 - 2A^3$
$A^3 \equiv A(x^3)$ and $A^2 \equiv x^2 + 2A(x^3) \pmod 3$. *(proved.)*
Paper · OEIS A389475 · Lean

## 7. Other results

Further entries, each with its own method.

**A396093.** When are the coefficients of $B(B(B(x)))$ odd?
$a(n)$ is odd exactly when $n \equiv 1, 3, 7, 9 \pmod{10}$. *(proved.)*
Paper · OEIS A396093 · Lean

**A396102.** Every coefficient is 1 modulo 3: the series with $A(A(A(x))) = (1 + x)\,A(A(x))$
$a(n) \equiv 1 \pmod 3$, together with a theorem for the whole family $F^{\circ r} = (1 + x)F^{\circ(r-1)}$. *(proved.)*
Paper · OEIS A396102 · Lean

**A380056.** The numbers with exponential generating function $(e^x - 1)/\cos 2x$, modulo 4 and 5
$a(n) \equiv 1 \pmod 4$, and $a(n) \equiv 2 \cdot 2^n + 3 \cdot 4^n \pmod 5$, so $5 \mid a(n)$ exactly when $4 \mid n$. *(proved.)*
Paper · OEIS A380056 · Lean

**A397346.** Residues modulo 8 for a series defined by a self-referential exponential

From $n = 2$ on, $a(n)$ mod 8 repeats $4, 2, 0, 2$; more precisely, $a(n)/n \equiv 6 \pmod 8$ if $n \equiv 3 \pmod 4$ and $a(n)/n \equiv 2 \pmod 8$ otherwise. *(proved.)*

Paper · OEIS A397346 · Lean

**A394576.** A greedy sequence that avoids sums of three consecutive terms

$\lfloor (4n-3)/3 \rfloor \le a(n) \le \lfloor (4n-1)/3 \rfloor$ for all $n \ge 1$; in particular $a(3k) = 4k - 1$. *(proved.)*

Paper · OEIS A394576 · Lean